Accelerated Math II/III Name_____

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HW #7 - Solving Polynomial Equations

Date_____ Period____

Factor each completely.

1) $35m^3 + 7m^2 - 20m - 4$

2) $48a^3 + 42a^2 - 56a - 49$

3) $96n^4 + 64n^3 - 120n^2 - 80n$

4) $20n^3 + 100n^2 - 15n - 75$

5) $8x^3 - 1$

6) $2u^3 + 432$

7) $192 - 81x^3$

8) $-108x^3 + 32$

9) $u^4 + 9u^2 + 14$

10) $2x^4 - 10x^2 + 8$

11) $14m^4 - 38m^2 - 12$

12) $7x^4 + 20x^2 - 3$

13) $8m^4 + 54m^2 + 81$

14) $60x^4 + 81x^2 - 24$

Find all roots. Write complex solutions in standard form (*a* + *ib*).

15) $x(5x - 1)(x + 1)^2 = 0$

16) $x(2x - 1)(x + 1)(x - 1) = 0$

17) $3x^3 + 19x^2 + 20x = 0$

18) $3x^3 - 5x^2 + 2x = 0$

19) $2x^4 + 11x^2 - 40 = 0$

20) $5x^4 - 6x^2 - 27 = 0$

21) $2x^4 + 2x^3 + 5x^2 + 5x = 0$

22) $2x^4 + 5x^3 + 6x^2 + 15x = 0$

23) $x^4 + 64x = 0$

24) $27x^4 - x = 0$

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Answers to HW #7 - Solving Polynomial Equations

1) $(7m^2 - 4)(5m + 1)$
2) $(6a^2 - 7)(8a + 7)$
3) $8n(4n^2 - 5)(3n + 2)$
4) $5(4n^2 - 3)(n + 5)$
5) $(2x - 1)(4x^2 + 2x + 1)$
6) $2(u + 6)(u^2 - 6u + 36)$
7) $3(4 - 3x)(16 + 12x + 9x^2)$
8) $4(-3x + 2)(9x^2 + 6x + 4)$
9) $(u^2 + 2)(u^2 + 7)$
10) $2(x - 2)(x + 2)(x - 1)(x + 1)$
11) $2(7m^2 + 2)(m^2 - 3)$
12) $(7x^2 - 1)(x^2 + 3)$
13) $(2m^2 + 9)(4m^2 + 9)$
14) $3(5x^2 + 8)(2x - 1)(2x + 1)$
15) $\left\{0, \frac{1}{5}, -1 \text{ mult. } 2\right\}$
16) $\left\{0, \frac{1}{2}, -1, 1\right\}$
17) $\left\{0, -\frac{4}{3}, -5\right\}$
18) $\left\{0, \frac{2}{3}, 1\right\}$
19) $\left\{\frac{\sqrt{10}}{2}, -\frac{\sqrt{10}}{2}, 2i\sqrt{2}, -2i\sqrt{2}\right\}$
20) $\left\{\frac{3i\sqrt{5}}{5}, -\frac{3i\sqrt{5}}{5}, \sqrt{3}, -\sqrt{3}\right\}$
21) $\left\{0, -1, \frac{i\sqrt{10}}{2}, -\frac{i\sqrt{10}}{2}\right\}$
22) $\left\{0, -\frac{5}{2}, i\sqrt{3}, -i\sqrt{3}\right\}$
23) $\left\{0, -4, 2 + 2i\sqrt{3}, 2 - 2i\sqrt{3}\right\}$
24) $\left\{0, \frac{1}{3}, \frac{-1 + i\sqrt{3}}{6}, \frac{-1 - i\sqrt{3}}{6}\right\}$

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